MINNESOTA STATE UNIVERSITY, MANKATO
DEPARTMENT OF COMPUTER INFORMATION SCIENCE
COMPUTER SCIENCE PROGRAM

HANDOVER DOCUMENT

THE MavPASS SCHEDULING PROJECT

Clients:
Dr. Laura Jacobi
MavPASS Faculty Liaison

Lina Wang
MavPASS Coordinator

THE MavPASS SCHEDULING TEAM

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Software Engineer

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PURPOSE

The purpose of this handover document is to give a detailed overview of all the major components of the MavPASS Scheduling project such as project approach, client deliverables, outcomes, and more. It is intended to help ensure that the project transitions smoothly from the first phase to the next.

PROJECT OVERVIEW

The Clients:

Dr. Laura Jacobi, the MavPASS Faculty Liaison.
More information on this client can be found on the MavPASS webpage at
https://www.mnsu.edu/academics/academic-support/student-support-centers/maverick-peer-facilitated-academic-support-system/about-mavpass2/mavpass-staffing/laura-jacobi/

Dr. Lina Wang, the MavPASS Coordinator.
More information on this client can be found on the MavPASS webpage at
https://www.mnsu.edu/academics/academic-support/student-support-centers/maverick-peer-facilitated-academic-support-system/about-mavpass2/mavpass-staffing/lina-wang/

About MavPASS

Implemented on the Minnesota State University Mankato campus in the fall of 2019, MavPASS is founded in supplemental instruction. It was branded as MavPASS because it is a "peer-facilitated academic support system" for students in historically difficult courses. To support students in these difficult courses, MavPASS Leaders are identified and hired to sit in on the course in which they already demonstrated proficiency in a previous semester. The MavPASS Leaders are also trained in collaborative learning techniques in order to facilitate study sessions for any students in participating sections of the course. The MavPASS sessions are confidential and free, and research shows that students who attend supplemental instruction regularly typically earn up to **one full letter grade higher** in the course.

Problem Case:

MavPASS, MSU Mankato's "Maverick Peer-Facilitated Academic Support System", runs a large and complex system of student support sessions each week of each semester. Over one hundred (100) sessions per week are commonly found on the schedule, serving thousands of students at a time.

The setup and operation of the schedule for the MavPASS program, which is core to its successful delivery of services, involves handling many interacting factors, including but not limited to:

- University course schedules in general
- University course schedules of classes with MavPASS support
- University group schedules (i.e. Athletics, Learning Communities, Student Government, etc.)
- MavPASS leader course schedules, work schedules, and other availability constraints
- Room availability
- Additional specific constraints such as desirability of back-to-back sessions

The MavPASS schedules are created using mostly manual manipulation of spreadsheets. The currently used techniques (which are well-documented) are time-consuming and labor-intensive, and thus expensive.

Purpose of Project:
The project's purpose is to eliminate the inefficient method used to schedule MavPASS sessions and to replace it with a speedy and efficient scheduling system.

Project Objectives:
The project's objective is to build a new (semi) automated scheduling system for MavPASS.

Business Case:
This is the best solution to the problem because it heavily reduces the manual input the clients need to do and the time it takes to create the schedules. It would free up the time of MavPASS staff to devote to training and other enhancements to the MavPASS program. More importantly, it would increase the likelihood that MavPASS sessions would be scheduled at prime times for leaders and students.

Goals and Metrics:
The success of this project will be determined by whether our software can adequately create optimal schedules, display them in desired formats, and possibly create schedules for upcoming semesters.

Expected Deliverables:
Client deliverables are:

- Requirements Analysis Document
- Data analysis and cleaning
- Design document
- Test document
- Working application, and
- User documentation

PROJECT SCOPE

IN SCOPE

The project's initial scope was defined around creating a full web app that would (semi) automate the scheduling process at MavPASS. The product would have consisted of three main components:

- Client-side (front-end)
- Application layer / Business logic, and
- Server-side (back-end)

However, the scope was renegotiated to produce software that would essentially achieve the same goal — (semi) automate the scheduling process at MavPASS — while still displaying these schedules in a visually appealing and easily comprehensible format (read: excel layouts) by room, day, and master schedule. This was done to redirect all focus and efforts toward developing code that would successfully tackle this complex scheduling problem which was at the minimum what the clients needed and what needed to be achieved to consider this project a successful one.

This meant that we cut any work having to do with the front-end — UI (User Interface) — from the scope to **focus solely on the business logic and the backend**. The UI, however, has been slated to be included in the scope of the second phase for the next team that picks this project up.

OUT OF SCOPE

In addition to the UI, it was originally thought of to try to connect to the MinnState system to automatically synchronize students' schedules with the product as and when changes were made to students' registered courses. But because that's a statewide system and would require way more resources and permissions than is available to us, we cannot access it, and as such, it was suggested to try to connect to MavConnect instead since it is local to our university. However, due to time constraints, this is a task the next team will have to take on.

One of our clients, Lina, also mentioned having access to some university scheduling system where she can view room availabilities for the semester. She requested looking into possibilities of interfacing with that system so the product automatically pulls room availabilities from there instead of having to manually populate the database with that data.

Our faculty coach, Lin, mentioned in conversation a task that could improve the codebase — modularizing the source code to generic functions to accommodate changing constraints. MavPASS constraints change from semester to semester, and it is likely scheduling criteria will change as time goes on. Thus, having a codebase that does not have to be entirely re-coded due to changing constraints is very important.

Work to optimize and test the source code began but due to time and resource constraints, this did not see completion. It is very important that the next team implements testing so that bugs could be identified and fixed as more features are added to the codebase. Additionally, it will be great if schedules could be generated much faster than they currently do at four to five minutes per schedule.

To summarize, at the minimum the next team will attempt to:

- Develop the UI for the product
- Modularize the codebase
- Further optimize the product's codebase
- Fully test the product
- Interface the product with MavConnect, the university's room scheduling system, and more.

Furthermore, a future work recommendation is to conduct a study of the limitations such as the possible upper bound of the schedule score and the implications of adding too many constraints (e.g., schedule infeasibility).

TIMELINE

The initial scope of our project included the backend, the business logic, and the front end. Thus, our initial timeline, Timeline A (see appendix), reflected that. However, as we dove into the work, we realized certain tasks were more complex than anticipated and required more time to be accomplished. Since we were now restricted by time constraints, we had to minimize the scope to focus on developing an algorithm that would create a feasible schedule and display them in desired formats. This meant allocating a lot more time to coding the algorithm.

We also wanted to use this product to help our clients schedule sessions for the upcoming Spring 2023 semester and thus included it in our new timeline. With these changes to the scope of our project, the timeline and milestones changed accordingly as seen in Timeline B in the appendix.

Again, due to time constraints, scheduling the Spring 2023 semester schedule has been slated to take place over the winter break. Our clients agreed to work with us during this timeframe.

DELIVERABLES

Within the renegotiated scope, we completed every — both medium and high — deliverables.

Deliverables	Type of work	Activities	Resources	Tech Skills	Priority
Requirements analysis document: meet with Laura and Lina (and possibly student MavPASS leaders) to understand clearly what they think a "winning" solution will be. Study and absorb the extensive documentation they have prepared that describes their current scheduling process. Document, in a simple and concise fashion, all the application requirements including data input, option/preference specification, and results output.	Requirements analysis and study of existing process. Documentation – in simple/concise format – of requirements for automated or semi-automated solution.	Requirements gathering meetings, study of existing scheduling process, drafting requirements document , getting feedback from clients and incorporating changes they request until agreement is reached.	Access to client leads, documentation existing process	Spreadsheets, word processing	High
User stories that capture and span intended use cases and requirements.	Writing user stories	Writing user stories and presenting them to client for feedback, making modifications as needed	Document creation tool to be agreed with client	Familiarity with user story approach and techniques	High
Acquire and clean full example sets of data to be used for development and testing. This is likely to include multiple sources.	Data analysis and data cleansing	Work with client to obtain and get approval on data sets to be used for development and testing	Access to client lead, data provided by client, analysis assistance from client	Data management	High
Wireframe mockup of user interface and user experience. Be sure to include how users will set up data input, specification of preferences and options, and, of course, delivery of resulting schedules.	UI mockup	Design and create wireframe version of user interface and full-application user experience, manage iterative feedback from client to achieve desired solution	Wireframe tool of team's choice	Use of selected wireframe tool	High
Design document for application. Special attention to be paid to designing the application so that it can be modified, updated, and maintained by client after the project is complete. (In other words, a full software lifecycle model must be included in the design document.)	Software design	Detailed design for data representation, scheduling algorithm, and user interface.	Online documentation for required development tools, as selected by project team and approved by client	Programming language and front end development tools to be chosen by team and approved by client. (Pay special attention to full lifecycle maintenance here.)	High
Test document for application.	Test planning, execution, and documentation	Creating test plan, executing tests iteratively, documenting results and managing incremental improvements.	Examples of test documents from other projects, support from IT team, word processing tools	Test planning, execution, and documentation	Medium
Working application.	Coding	Implement Front End, Server layer, and Database to create full stack application.	See above	Programming language and front end development tools as previously agreed.	Medium
Detailed user documentation that includes all steps needed to make the application work as each new semester arrives.	Documentation	Documenting all steps the client user must take to adapt the application as each new semester arrives.	Close interaction with client leads, acceptance of client leads	Word processing	High

Figure A: Excerpt from the MavPASS Scheduling MSU CS Project Description and Deliverables document.

The key technical work involved in completing the first deliverable was:

- Requirements analysis
  The team held meetings with the clients, Laura and Lina, and some student MavPASS leaders over a week to understand clearly what they thought a "winning" solution would be. Based on their initial responses, we asked clarifying questions and made suggestions predicated on technical expertise that drove further insightful conversations. This enabled us to accurately nail down exactly what they wanted the product to do.

- Study of the existing system.
  While they did not have documentation on how their current scheduling process is conducted, we found out through discussions what this was and formally documented it. We then carefully studied it and drew from it information that contributed to key decision-making processes.

After conducting both of these activities, we documented, simply and concisely, all the application requirements including data input, option/preference specification, and results output. This is detailed in the requirements analysis document that was submitted to the clients. They made some changes to the document and then communicated approval which gave us the go-ahead to begin working on the other deliverables.

Four of the six deliverables in scope could be classified under the two components of the product:

Back-end
After analyzing the data given to us by the clients, discussing the pros and cons of using a relational database model versus a non-relational database model, and in addition, consulting a subject matter expert, Dr. Jonathan Hardwick, we decided to implement a relational database model.

The two reasons for the dilemma that led to the aforementioned discussion were:

i. while a relational database model appeared to be most suitable since the data provided was mostly structured, non-relational databases are more scalable since MavPASS is a growing program with increasingly more courses being supported and thus more leaders being hired leading to larger datasets.
ii. we easily came up with a solution (read: data structure) for how to represent our schedule solutions in NoSQL while it was proving difficult to do the same in SQL.

However, we concluded that MavPASS datasets are not growing fast and large enough to want to implement database solutions in NoSQL and that while it would take a bit longer to figure out how to represent our schedule solutions in SQL, it was the better way to go since we were likely to run into issues later on when we eventually began implementing the solutions with NoSQL.

Key technical work involved in implementing the database included:

- Analyzing and cleaning the data
  In conducting analyses of the data, we realized it contained a lot of inconsistencies and inaccuracies. We cross-checked with eservices and the MavPASS page on the school website (https://www.mnsu.edu/academics/academic-support/student-support-centers/maverick-peer-facilitated-academic-support-system/mavpass-courses/) and consulted and worked with our clients to correct this. We then transformed the cleaned data into formats that could be easily processed by our algorithm. One such format is the bit-string representation of leader and room availability where 0 indicates unavailability and 1, availability. This was stored in an excel book.

- Designing and developing the database
  After analyzing and cleaning the data, we broke it down into entities (a thing, place, person, unit, object, or any item about which data can be stored) and detailed the relationships that existed between them. We then designed a database model based on that and sought feedback from a subject matter expert, Dr. John Burke, who gave us feedback after every improved design. It was an iterative design process that involved a lot of database normalization which eventually led to an optimum design. This was documented in the database design document. After receiving the go-ahead on our final design, we developed the database in MySQL and populated it with our cleaned and formatted data.

Application layer
Genetic algorithms were adapted to develop a solution that would become the business logic.
The steps involved in genetic algorithms are:

- Creating a genome (a solution schedule)
- Generating a population (the set of solution schedules that meet the hard constraints)
- Evaluating the fitness (based on soft constraints)
- Mutating the schedule (changing a single piece of information and re-evaluating the fitness which in this case was changing some soft constraints to hard constraints)

After a lot of brainstorming, the team came up with the following solution to generate a genome and population:

1. Randomly pick a course from the list of supported courses

2. Randomly pick a leader from the list of leaders supporting that course
3. Randomly pick an available time from the leader's bit-string availability
4. Randomly pick a room from a list of rooms that are available at the randomly picked leader's availability time
5. Schedule the leader for a session at that time
6. Count the number of unscheduled sessions left for that leader and then repeat steps 3-6 until all sessions have been scheduled for that leader.
7. Remove that leader from the list of leaders supporting that course and repeat steps 2-7.
8. Do this until all leaders for that course have been scheduled then remove that course from the list of supported courses and repeat steps 1-8.
9. Do this until all leaders across all courses have been scheduled.

Key technical work involved in implementing this solution included:

- Coding the algorithm
  We implemented the above algorithm using object-oriented programming. Our well-commented code files explain what part each plays in executing the above solution. In order to ensure that we were not producing bad schedules, truth tables were created from the Important Scheduling Notes (ISN) document to be used as metrics in checking how good the schedules produced were. This is documented in the Fitness Scores document. The schedules produced were then scored based on those truth tables to measure how best they fit the scheduling criteria i.e., evaluating the fitness. The score obtained after executing the first schedule was low and that informed our decision to move the ISN 4a soft constraint to a hard constraint. This drastically improved the schedule score which implied that sessions were now being scheduled at the best times considering when courses met for lectures. The main objective of improving the score, from 63% to 88%, was to produce schedules that better met the scheduling criteria and improved on the current semester's schedule which was scored at 76% (see Fall 2022 Schedule Scoring).

- Testing
  A detailed test plan documenting what tests to implement, how to execute tests iteratively, document results, and manage incremental improvements was created. However, due to time and resource constraints, only minimal testing was conducted. We developed a few unit tests for the base functions in our algorithm to ensure that the product was working as expected.

The last deliverable involved creating detailed user documentation that included all steps needed to make the application work. Paying heed to user and client sensitivity, we clearly outlined the simplest way — which required little to no technical expertise or background — to run the software. We, however, also offered to troubleshoot should they run into any issues running the product.

LEARNING, RESOURCES, AND TOOLS

Since every member of the team had varying levels of domain knowledge, the primary learning objective of this project was to get acquainted with databases, C#, testing, version control, and genetic algorithms so as to be able to apply these computer science theories and software development fundamentals to produce a computing-based solution. (CS graduation and ABET outcome 6)

We learned most of these by watching YouTube and Coursera videos, teaching each other, and reading from shared textbooks, and links to online resources.

For this project, we used MySQL, Visual Studio 2022 (a C# IDE), nUnit testing framework, and GitHub.

MySQL can be downloaded at https://www.mysql.com/downloads/
MySQL Workbench can be downloaded at https://www.mysql.com/products/workbench/
Visual Studio can be downloaded at https://visualstudio.microsoft.com/downloads/
GitHub repository can be found at https://github.com/MNSU-MavPASS/MavPASS_Scheduling

The MySQL server can be accessed with the host named MavPASS Scheduling. The port, username, and password will be omitted for confidentiality and security reasons.

REFERENCES

These refer to all the documents that were used in the entirety of the project for some task or another. The document that first contextualized the essence of the project was made available to us by our faculty coach in advance of the semester: MavPASS Scheduling MSU CS Project Description and Deliverables.

CLIENT-PROVIDED DOCUMENTS

At the beginning of the semester, our clients provided us with the data needed to model our project for development and testing purposes. This data was what the clients used to draw up the schedule currently in use for the Fall 2022 semester. It contains room availabilities, relevant information on leaders and their availabilities, and a color-coded bar graph of said availabilities to easily spot when leaders are free, willing, and able to conduct MavPASS sessions and/or attend training meetings.

Halfway through the semester, they also provided us with a document detailing the criteria by which they usually schedule sessions and leaders to rooms. This document additionally includes information on how rooms are assigned to certain courses based on enrollment and attendance data.

Listed below are the documents:

- Fall 2022 Data (zipped file)
    - Available Times for MP Sessions F22
    - F22 MavPASS Leaders Availability Bar Graph F22
    - FH 216 Sessions F22
    - ML86 Sessions F22
    - ML47 Sessions F22
    - ML101B Sessions F22
    - ML109 Sessions F22
    - ML1034 Session F22
    - PS105 Sessions F22
- Important Scheduling Notes (ISN)

PROJECT-DEVELOPED DOCUMENTS

The following documents and code were produced in line with the key technical work done and the deliverables expected of us.

Listed below are the documents:

- **Cleaned and formatted Datasets**: contains transformed sample data.

- **Requirements Analysis Document**: outlines the requirements for the MavPASS Scheduling product that need to be met for the project to be considered successful. It also documents the findings after careful analyses of the clients' current system were undertaken.

- **Database Design Document**: describes the design of the MavPASS Scheduling App database and maps the logical data model to the target Database Management System (DBMS), MySQL. It outlines how the data driving the app must be stored and how the data elements interrelate.

- **Test Plan Document**: describes the testing approach and overall framework that will be used to ensure testing activities are effectively implemented in the MavPASS Scheduling Project.

- **MavPASS_Scheduling Code (zipped file)**: contains code files that implement the business logic.

- **Fitness Scores Document**: details the scoring metrics based on scheduling criteria.

- **Fall 2022 Schedule Scoring**: details the score breakdown and overall score of the MavPASS schedule for the current Fall 2022 semester according to MavPASS scheduling criteria.

- **User Documentation**: describes the product developed by the MavPASS Scheduling Team for its end-users.

APPENDICES

Timeline A

Name	Start date	Internal Due	External Due Date	Type	Description		Completion	Status
Project Duration	@August 24, 2022	@December 2, 2022	@December 6, 2022	School Deliverable	Timeframe within which to work on the project	High		In progress
Light Learning	@August 24, 2022	@December 2, 2022	@December 6, 2022	Internal Team Deliverable	Learning that is designed to take less time	Medium		In progress
Access to Client database	@August 24, 2022	@August 26, 2022	@August 28, 2022	Internal Team Deliverable	Acquiring access to the existing database and operational documention	High	@August 26, 2022	Completed
Coach Meeting Plan	@August 24, 2022	@August 29, 2022	@August 30, 2022	School Deliverable	Schedule for when and where to meet our coaches and subject matter experts	High	@August 27, 2022	Completed
Project Learning Plan	@August 24, 2022	@August 30, 2022	@August 31, 2022	School Deliverable	The what, how, and where to acquire the necessary skills and knowledge needed to complete this project	High	@August 26, 2022	Completed
Team Contract	@August 24, 2022	@August 31, 2022	@September 2, 2022	School Deliverable	Contract between team members detailing individual roles, expectations, and ground rules for better	High	@August 31, 2022	Completed
Project Timeline and Milestones	@August 26, 2022	@September 2, 2022	@September 7, 2022	School Deliverable	A guide listing out, in chronological order, tasks and estimated timeframes within which to complete them	High		Completed
Requirement Analysis Document	@September 6, 2022	@September 12, 2022	@September 16, 2022	Client Deliverable	Document detailing user expectations for the program about to be built	High		In progress
Tools Setup	@September 7, 2022	@September 11, 2022	@September 11, 2022	Internal Team Deliverable	Finish setting up all the tools and environment in our chosen tech stack	High		In progress
1Q Term Project Review	@September 7, 2022	@September 12, 2022	@September 14, 2022	School Deliverable	Internal status update on project progress. An in-person information sharing and feedback session	High		In progress
Data Analysis and Cleansing	@September 9, 2022	@September 12, 2022	@September 14, 2022	Client Deliverable	Study and clean the datasets obtained to be used for development and testing	High		Not started
Heavy Learning	@September 12, 2022	@September 25, 2022	@September 25, 2022	Internal Team Deliverable	Learning that's designed around taking extensive time. Includes tech talks where we	High		Not started
Test Plan Document	@September 16, 2022	@September 23, 2022	@September 25, 2022	Client Deliverable	Detailed document that catalogs the test strategy, objectives, schedule,	Medium		Not started
Coding the Scheduling Algorithm	@September 21, 2022	@October 5, 2022	@October 7, 2022	Client Deliverable	Development of source code that will solve and automate the scheduling problem the clients have presented to us.	High		Not started
Unit Testing	@September 21, 2022	@October 5, 2022	@October 7, 2022	Client Deliverable	Testing samllest, individual units of code to ensure base/source code is	High		Not started
Creating and Populating the Database	@September 26, 2022	@September 28, 2022	@September 28, 2022	Client Deliverable	Development and feeding of relational database with clean data	High		Not started
Midterm Evaluation Presentation	@October 7, 2022	@October 10, 2022	@October 12, 2022	School Deliverable	Presentation to faculty and coaches on progress made. Showcasing in general what we have done and what more	High		Not started

Midterm Project Review Materials Uploaded	@October 7, 2022	@October 15, 2022	@October 15, 2022	School Deliverable	Submit project presentation contents. Including project artifacts supporting materials referred to and/or used in presentation	High		Not started
User Stories	@October 10, 2022	@October 14, 2022	@October 16, 2022	Client Deliverable	Informal, natural language description of features of a	High		Not started
UI Mockup [Wireframes]	@October 14, 2022	@October 19, 2022	@October 21, 2022	Client Deliverable	Schematic or blueprint that is useful for helping	High		Not started
Design Documentation	@October 14, 2022	@October 21, 2022	@October 23, 2022	Client Deliverable	Documents a complete high-level solution to	High		Not started
Coding Frontend	@October 14, 2022	@November 22, 2022	@November 22, 2022	Client Deliverable	Development of the graphical user interface of a	Medium		Not started
Code and Documentation Review #1	@October 15, 2022	@October 19, 2022	@October 21, 2022	School Deliverable	Round-robin style mutual review of others' project documentation and code	High		Not started
3Q Term Project Review	@November 1, 2022	@November 7, 2022	@November 9, 2022	School Deliverable	Internal status update on project progress	High		Not started
Code and Documentation Review #2	@November 6, 2022	@November 13, 2022	@November 18, 2022	School Deliverable	Round-robin style mutual review of others' project documentation and code	High		Not started
User Testing	@November 11, 2022	@November 16, 2022	@November 18, 2022	Client Deliverable	Getting clients to test and evaluate the product	High		Not started
Connecting Back-end to the Front-End	@November 14, 2022	@November 22, 2022	@November 22, 2022	Client Deliverable	Integrating client side with server side of the product	High		Not started
Performance Review	@November 16, 2022		@November 19, 2022	School Deliverable	Review how the team in general and everyone	High		Not started
Handover Report	@November 21, 2022	@November 30, 2022	@December 3, 2022	School Deliverable	Document that is designed to contextualize all	High		Not started
Code Freeze	@November 22, 2022	@September 22, 2022	@November 22, 2022	School Deliverable	Halt code development and focus on testing	High		Not started
Quality Assurance	@November 22, 2022	@November 30, 2022	@December 2, 2022	Client Deliverable Internal Team Deliverable	Maintenance efforts taken to ensure that the product delivered	High		Not started
User Documentation	@November 22, 2022	@November 30, 2022	@December 3, 2022	Client Deliverable	Document designed for end-users containing information describing the use,	High		Not started
Final Project Package	@November 22, 2022	@December 2, 2022	@December 3, 2022	School Deliverable	Combine, package up and submit all elements of	High		Not started
Product Presentation to Client	@November 22, 2022	@December 6, 2022	@December 8, 2022	Client Deliverable	Final delivery to client	High		Not started
Final Project Presentation	@November 25, 2022	@December 1, 2022	@December 6, 2022	School Deliverable	Final internal presentation on what was	High		Not started
Final Project Presentation Materials Uploaded	@November 28, 2022	@December 2, 2022	@December 6, 2022	School Deliverable	Submit final project presentation contents.	High		Not started

Timeline B

Name	Start date	Internal Due Date	External Due ...	Completion Date	Type	Description	Priority	Status
Project Learning Plan	August 24, 2022	August 30, 2022	August 31, 2022	August 26, 2022	School Deliverable	The what, how, and where to acquire the nece	High	Completed
Access to Client database	August 24, 2022	August 26, 2022	August 28, 2022	August 26, 2022	Internal Team Delive	Acquiring access to the existing database and	High	Completed
Coach Meeting Plan	August 24, 2022	August 29, 2022	August 30, 2022	August 27, 2022	School Deliverable	Schedule for when and where to meet our co	High	Completed
Team Contract	August 24, 2022	August 31, 2022	September 2, 2022	August 31, 2022	School Deliverable	Contract between team members detailing in	High	Completed
Project Timeline and Milestones	August 26, 2022	September 2, 2022	September 7, 2022	September 6, 2022	School Deliverable	A guide listing out, in chronological order, tas	High	Completed
Tools Setup	September 7, 2022	September 11, 2022	September 11, 2022	September 8, 2022	Internal Team Delive	Finish setting up all the tools and environmer	High	Completed
Requirement Analysis Document	September 6, 2022	September 12, 2022	September 16, 2022	September 12, 2022	Client Deliverable	Document detailing user expectations for the	High	Completed
1Q Term Project Review	September 7, 2022	September 12, 2022	September 14, 2022	September 14, 2022	School Deliverable	Internal status update on project progress. Ar	High	Completed
GitHub & JIRA Standards Demo		September 21, 2022		September 21, 2022	Internal Team Delive	A demo given by Madeline covering our GitH	Medium	Completed
Documentation Standards Demo		September 23, 2022		September 23, 2022	Internal Team Delive	A demo given by Madeline covering our Docu	Medium	Completed
Finalize Tech Stack	September 7, 2022	September 7, 2022	September 7, 2022	September 27, 2022	Internal Team Delive	Finalize what tools we will be using	High	Completed
Data Analysis and Cleansing	September 9, 2022	October 6, 2022	September 8, 2022	October 5, 2022	Client Deliverable	Study and clean the datasets obtained to be u	High	Completed
Creating and Populating the Databas	September 26, 2022	October 11, 2022	October 11, 2022	October 10, 2022	Client Deliverable	Development and feeding of relational datab	High	Completed
Test Plan Document	September 16, 2022	October 10, 2022	October 10, 2022	October 10, 2022	Client Deliverable	Detailed document that catalogs the test stra	Medium	Completed
Midterm Evaluation Presentation	October 7, 2022	October 10, 2022	October 12, 2022	October 12, 2022	School Deliverable	Presentation to faculty and coaches on progr	High	Completed
User Stories	October 10, 2022	October 14, 2022	October 16, 2022	October 14, 2022	Client Deliverable	Informal, natural language description of feat	High	Completed
Midterm Project Review Materials Up	October 7, 2022	October 15, 2022	October 15, 2022	October 15, 2022	School Deliverable	Submit project presentation contents. Includi	High	Completed
Design Documentation	October 14, 2022	October 21, 2022	October 23, 2022	October 20, 2022	Client Deliverable	Documents a complete high-level solution to	High	Completed
Code and Documentation Review #1	October 15, 2022	October 19, 2022	October 21, 2022	October 20, 2022	School Deliverable	Round-robin style mutual review of others' pr	High	Completed
3Q Term Project Review	November 1, 2022	November 7, 2022	November 9, 2022	November 7, 2022	School Deliverable	Internal status update on project progress	High	Completed
Code and Documentation Review #2	November 6, 2022	November 13, 2022	November 18, 2022	November 13, 2022	School Deliverable	Round-robin style mutual review of others' pr	High	Completed
Unit Testing	September 21, 2022	November 22, 2022	November 22, 2022	November 22, 2022	Client Deliverable	Testing smallest, individual units of code to e	High	Completed
Coding the Scheduling Algorithm	September 21, 2022	November 22, 2022	November 22, 2022	November 22, 2022	Client Deliverable	Development of source code that will solve a	High	Completed
Code Freeze	November 22, 2022	November 22, 2022	November 22, 2022	November 28, 2022	School Deliverable	Halt code development and focus on testing	High	Completed
Collect Data to Generate a schedule f	November 21, 2022	November 30, 2022	December 3, 2022	November 30, 2022	Client Deliverable	we will collect course data, leader data with a	High	Completed
Quality Assurance	November 22, 2022	November 30, 2022	December 2, 2022	November 30, 2022	Client Deliverable	Maintenance efforts taken to ensure that the	High	Completed
Light Learning	August 24, 2022	December 2, 2022	December 6, 2022	December 2, 2022	Internal Team Delive	Learning that is designed to take less time	Medium	Completed
Product Presentation to Client	November 22, 2022	December 1, 2022	December 3, 2022	December 3, 2022	Client Deliverable	Final delivery to client	High	Completed
Final Project Package OPEN	November 22, 2022	December 2, 2022	December 3, 2022	December 3, 2022	School Deliverable	Combine, package up and submit all element	High	Completed
User Documentation	November 22, 2022	November 30, 2022	December 3, 2022	December 3, 2022	Client Deliverable	Document designed for end-users containing	High	Completed
Heavy Learning	September 12, 2022	December 3, 2022	December 6, 2022	December 3, 2022	Internal Team Delive	Learning that's designed around taking exten	High	Completed
Handover Report	November 21, 2022	December 3, 2022	December 3, 2022	December 3, 2022	School Deliverable	Document that is designed to contextualize a	High	Completed
Project Duration	August 24, 2022	December 2, 2022	December 6, 2022	December 3, 2022	School Deliverable	Timeframe within which to work on the proje	High	In progress
Final Project Presentation Materials U	November 28, 2022	December 2, 2022	December 6, 2022	December 6, 2022	School Deliverable	Submit final project presentation contents.	High	Completed
Final Project Presentation	November 25, 2022	December 5, 2022	December 6, 2022	December 6, 2022	School Deliverable	Final internal presentation on what was accon	High	Completed
Final Project Presentation Materials	December 2, 2022	December 5, 2022	December 6, 2022	December 7, 2022	School Deliverable		High	In progress
Performance Review	November 16, 2022	November 19, 2022	November 19, 2022	December 19, 2022	School Deliverable	Review how the team in general and everyon	High	Completed
Generate schedule for the next semes	November 24, 2022	November 19, 2022	November 30, 2022	December 31, 2022	Client Deliverable	use the product we developed to try and pro	High	In progress
User Testing	November 11, 2022	November 16, 2022	November 18, 2022		Client Deliverable	Getting clients to test and evaluate the produ	High	Not started